

2520090235_Practical4

The screenshot shows the MySQL Workbench interface. The SQL Editor contains the following script:

```
1 • drop database if exists bookflow_db;
2 • CREATE DATABASE bookflow_db;
3 • USE bookflow_db;
4 • SELECT DATABASE();
```

The Output tab shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	14:37:18	Drop database if exists bookflow_db	Error Code: 1064. You have an error in your SQL synta...	0.000 sec
2	14:37:35	drop database if exists bookflow_db	8 row(s) affected	0.203 sec
3	14:37:46	CREATE DATABASE bookflow_db	1 row(s) affected	0.000 sec
4	14:37:54	USE bookflow_db	0 row(s) affected	0.000 sec
5	14:38:31	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

The left sidebar shows the 'MANAGEMENT' and 'INSTANCE' sections. The 'Information' tab is active, showing 'No object selected'. The bottom status bar indicates 'Query Completed'.

MySQL Workbench

MySQL Model xLocal instance MySQL80 x

FileEditViewQueryDatabaseServerToolsScriptingHelp

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs

AdministrationSchemas

Information

No object selected

Week_1_practicalSQL File 3*SQL File 4*

Limit to 1000 rows

SQLAdditions

13genre VARCHAR(100),

14

15description TEXT,

16

17published_year INT

18};

19describe Books;

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Result Grid

Form Editor

FieldTypeNullKeyDefaultExtra

book_idintNONOPRINULLauto_increment

titlevarchar(200)NONONULL

authorvarchar(150)NONONULL

genrevarchar(100)YESNONULL

descriptiontextYESNONULL

published_yearintYESNONULL

Read OnlyContext HelpSnippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
2	14:37:35	drop database if exists bookflow_db	8 row(s) affected	0.203 sec
3	14:37:46	CREATE DATABASE bookflow_db	1 row(s) affected	0.000 sec
4	14:37:54	USE bookflow_db	0 row(s) affected	0.000 sec
5	14:38:31	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
6	14:39:13	CREATE TABLE Books (book_id INT AUTO_INC...	0 row(s) affected	0.047 sec
7	14:39:55	describe Books	6 row(s) returned	0.015 sec / 0.000 sec

Object InfoSession

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Workbench

MySQL Model x Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator Week_1_practical SQL File 3* SQL File 4* x SQLAdditions

MANAGEMENT

- Server Status
- Client Connections
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- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs

Administration Schemas

Information

No object selected

```

112 author,
113 genre,
114 published_year
115 FROM Books
116 ORDER BY book_id;
117
118

```

Result Grid Filter Rows: Edits: Export/Import: Result Grid Form Editor

book_id	title	author	genre
1	The Martian	Andy Weir	Science Fiction
2	Dune	Frank Herbert	Science Fiction
3	The Hitchhikers Guide to the Galaxy	Douglas Adams	Science Fiction
4	Pride and Prejudice	Jane Austen	Romance
5	The Great Gatsby	F. Scott Fitzgerald	Classic
6	Foundation	Isaac Asimov	Science Fiction
7	The Hobbit	J.R.R. Tolkien	Fantasy

Books 3 x Apply Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
4	14:37:54	USE bookflow_db	0 row(s) affected	0.000 sec
5	14:38:31	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
6	14:39:13	CREATE TABLE Books (book_id INT AUTO_INC...	0 row(s) affected	0.047 sec
7	14:39:55	describe Books	6 row(s) returned	0.015 sec / 0.000 sec
8	14:40:52	INSERT INTO Books (title, author, genre, ...	10 row(s) affected Records: 10 Duplicates: 0 Wami...	0.032 sec
9	14:41:19	SELECT book_id, title, author, genre, pu...	10 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Query Completed

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MySQL Workbench

MySQL Model

Local instance MySQL80

FileEditViewQueryDatabaseServerToolsScriptingHelp

Navigator

Week_1_practicalSQL File 3*SQL File 4*

SQLAdditions

MANAGEMENT

Server Status

Client Connections

Users and Privileges

Status and System Va

Data Export

Data Import/Restore

INSTANCE

Startup / Shutdown

Server Logs

Administration

Information

No object selected

Limit to 1000 rows

Jump to

113genre,

114published_year

115FROM Books

116ORDER BY book_id;

117ALTER TABLE Books

118ADD COLUMN embedding JSON;

119DESCRIBE Books;

120

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Result Grid

Form Editor

Result 4

Read Only

Context Help

Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
9	14:41:19	SELECT book_id, title, author, genre, publi...	10 row(s) returned	0.000 sec / 0.000 sec
10	14:43:29	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec
11	14:44:04	DESCRIBE Books	7 row(s) returned	0.000 sec / 0.000 sec

Object Info

Session

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Workbench

MySQL Model x Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator Week_1_practical SQL File 3* SQL File 4* SQLAdditions

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Va
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- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs

Administration

Information

No object selected

Result Grid Filter Rows: Edit: Export/Import: Result Grid Form Editor

```
149 WHERE book_id = 10;
150 SELECT
151     book_id,
152     title,
153     embedding
154 FROM Books
155 ORDER BY book_id;
156
```

book_id	title	embedding
1	The Martian	[0.90, 0.80, 0.20]
2	Dune	[0.85, 0.75, 0.25]
3	The Hitchhikers Guide to the Galaxy	[0.80, 0.70, 0.30]
4	Pride and Prejudice	[0.10, 0.20, 0.90]
5	The Great Gatsby	[0.15, 0.30, 0.80]
6	Foundation	[0.95, 0.65, 0.20]
7	The Hobbit	[0.30, 0.75, 0.90]
8	Enders Game	[0.88, 0.85, 0.15]

Books 5 x Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 20	14:50:57	UPDATE Books SET embedding = JSON_ARRAY(0.20...	1 row(s) affected Rows matched: 1 Changed: 1 Wami...	0.015 sec
✓ 21	14:51:08	UPDATE Books SET embedding = JSON_ARRAY(0.70...	1 row(s) affected Rows matched: 1 Changed: 1 Wami...	0.016 sec
✓ 22	14:51:19	SELECT book_id, title, embedding FROM Book...	10 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Workbench

MySQL Model x Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: Week_1_practical SQL File 3* SQL File 4* x SQLAdditions

MANAGEMENT

- Server Status
- Client Connections
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INSTANCE

- Startup / Shutdown
- Server Logs

Administration

No object selected

Information

Result Grid

	book_id	title	author	genre	l2_distance
1	1	The Martian	Andy Weir	Science Fiction	0
8	8	Enders Game	Orson Scott Card	Science Fiction	0.0735
2	2	Dune	Frank Herbert	Science Fiction	0.0866

Result 6 x Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
21	14:51:08	UPDATE Books SET embedding = JSON_ARRAY(0.70...	1 row(s) affected Rows matched: 1 Changed: 1 Wami...	0.016 sec
22	14:51:19	SELECT book_id, title, embedding FROM Book...	10 row(s) returned	0.000 sec / 0.000 sec
23	14:56:26	SELECT book_id, title, author, genre, R...	3 row(s) returned	0.016 sec / 0.000 sec

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator: Week_1_practical SQL File 3* SQL File 4*

MANAGEMENT

- Server Status
- Client Connections
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INSTANCE

- Startup / Shutdown
- Server Logs

Administration

Information: No object selected

Result Grid

book_id	title	author	genre	cosine_similarity
1	The Martian	Andy Weir	Science Fiction	1
2	Dune	Frank Herbert	Science Fiction	0.9986
8	Enders Game	Orson Scott Card	Science Fiction	0.9983

Result 7 x Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
22	14:51:19	SELECT book_id, title, embedding FROM Books	10 row(s) returned	0.000 sec / 0.000 sec
23	14:56:26	SELECT book_id, title, author, genre, cosine_similarity FROM Books	3 row(s) returned	0.016 sec / 0.000 sec
24	15:03:59	SELECT book_id, title, author, genre, cosine_similarity FROM Books	3 row(s) returned	0.000 sec / 0.000 sec

Query Completed

Explorer

BOOKFLOW_API

- node_modules
- package-lock.json
- package.json

package.json

```
{
  "name": "bookflow-api",
  "version": "1.0.0",
  "main": "server.js",
  "scripts": {
    "start": "node server.js"
  },
  "dependencies": {
    "express": "^5.1.0",
    "mysql2": "^3.14.3"
  }
}
```

Terminal

```
PS C:\Users\snigd\Bookflow_API> npm install

added 78 packages, and audited 79 packages in 9s

29 packages are looking for funding
  run 'npm fund' for details

found 0 vulnerabilities
```

The image shows a Visual Studio Code editor window with a project named 'Bookflow_API'. The Explorer sidebar on the left shows the file structure: 'node_modules', 'package-lock.json', 'package.json', and 'server.js'. The 'server.js' file is open in the editor, showing the following code:

```
1 const express = require('express');
2 const mysql = require('mysql2/promise');
3
4 const app = express();
5 const PORT = 3000;
6
7 app.use(express.json());
8
9 const pool = mysql.createPool({
10   host: 'localhost',
11   user: 'root',
12   password: 'sumanjali@123',
13   databasd: 'bookflow_id',
14   waitForConnections: true,
15   connectionLimit: 10,
16   queueLimit: 0
17 });
18
19 app.get('/', (req, res) => {
20   res.json()
```

Below the editor, the 'Terminal' tab is active, showing the following commands and output:

```
PS C:\Users\sngid\Bookflow_API> npm install
29 packages are looking for funding
  run 'npm fund' for details

found 0 vulnerabilities
PS C:\Users\sngid\Bookflow_API> node server.js
BookFlow API running at http://localhost:3000
```

The status bar at the bottom indicates 'Ln 8, Col 1', 'Spaces: 4', 'UTF-8', 'CRLF', and 'JavaScript'.

The image shows a web browser window with the address bar set to 'localhost:3000'. The page content displays a JSON response from the API, which has been formatted using the 'Pretty-print' feature. The response is as follows:

```
{
  "message": "BookFlow Intelligent Catalog API",
  "status": "Running",
  "endpoints": {
    "search": "GET /v1/books/search",
    "createBook": "POST /v1/books"
  }
}
```

Swagger Editor
Powered by SMART2840

File

Edit

Generate Server

Generate Client

About

Sign in to Swagger

Try Swagger

1openapi: 3.0.3

2

3info:

4 title: BookFlow Intelligent Catalog API

5 version: 1.0.0

6 description: |

7 REST API for semantic book search.

8

9 The API searches books using mock

10 three-dimensional vectors stored in MySQL.

11

12servers:

13 - url: http://localhost:3000

14

15tags:

16 - name: Books

17 description: Book operations

18

19paths:

20

21 /v1/books/search:

22 get:

23 tags:

24 - Books

25

26 summary:

27 Search books using semantic similarity

28

29 parameters:

30

31 - name: q

BookFlow Intelligent Catalog API

1.0.0OAS 3.0

REST API for semantic book search.

The API searches books using mock three-dimensional vectors stored in MySQL.

Servers

http://localhost:3000

Books

Book operations

GET

/v1/books/searchSearch books using semantic similarity

POST

/v1/booksCreate a new book

Schemas

Curl

curl -X 'GET' \
 'http://localhost:3000/v1/books/search?q=Stories%20about%20space&limit=3' \
 -H 'accept: application/json'

Request URL

http://localhost:3000/v1/books/search?q=Stories%20about%20space&limit=3

Server response

Code

Details

200

Response body

```
{
  "description": "An astronaut struggles to survive alone on Mars.",
  "published_year": 2011,
  "similarity": 1
},
{
  "book_id": 2,
  "title": "Dune",
  "author": "Frank Herbert",
  "genre": "Science Fiction",
  "description": "A story involving planets, politics, survival and space.",
  "published_year": 1965,
  "similarity": 0.9986
},
{
  "book_id": 8,
  "title": "Enders Game",
  "author": "Orson Scott Card",
  "genre": "Science Fiction",
  "description": "A young student is trained for an interstellar conflict."
}
```

Swagger Editor

File Edit Generate Server Generate Client About

Sign in to Swagger Try Swagger

```
1 openapi: 3.0.3
2
3 info:
4   title: BookFlow Intelligent Catalog API
5   version: 1.0.0
6   description: |
7     REST API for semantic book search.
8
9   The API searches books using mock
10  three-dimensional vectors stored in MySQL.
11
12 servers:
13   - url: http://localhost:3000
14
15 tags:
16   - name: Books
17     description: Book operations
18
19 paths:
20   /v1/books/search:
21     get:
22       tags:
23         - Books
24       summary:
25         Search books using semantic similarity
26       parameters:
```

Request URL

http://localhost:3000/v1/books

Server response

Code Details

201

Response body

```
{
  "message": "Book created successfully",
  "book": {
    "book_id": 11,
    "title": "Project Hail Mary",
    "author": "Andy Weir",
    "genre": "Science Fiction",
    "description": "A science fiction survival story involving spac...",
    "published_year": 2021,
    "embedding": [
      0.03,
      0.02,
      0.12
    ]
  }
}
```

Response headers

```
content-length: 253
content-type: application/json; charset=utf-8
```

Responses

Code	Description	Links
201	Book created successfully	No links

MySQL Workbench

MySQL Model x Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

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INSTANCE

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- Server Logs

Administration

Information

No object selected

Week_1_practical SQL File 3* SQL File 4*

Limit to 1000 rows

```
311 SELECT
312     book_id,
313     title,
314     author,
315     genre,
316     published_year
317 FROM Books
318 WHERE title = 'Project Hail Mary';
319
```

Result Grid

book_id	title	author	genre	published_year
11	Project Hail Mary	Andy Weir	Science Fiction	2021
NULL	NULL	NULL	NULL	NULL

Books 8 x Apply Revert Context Help Snippets

Output

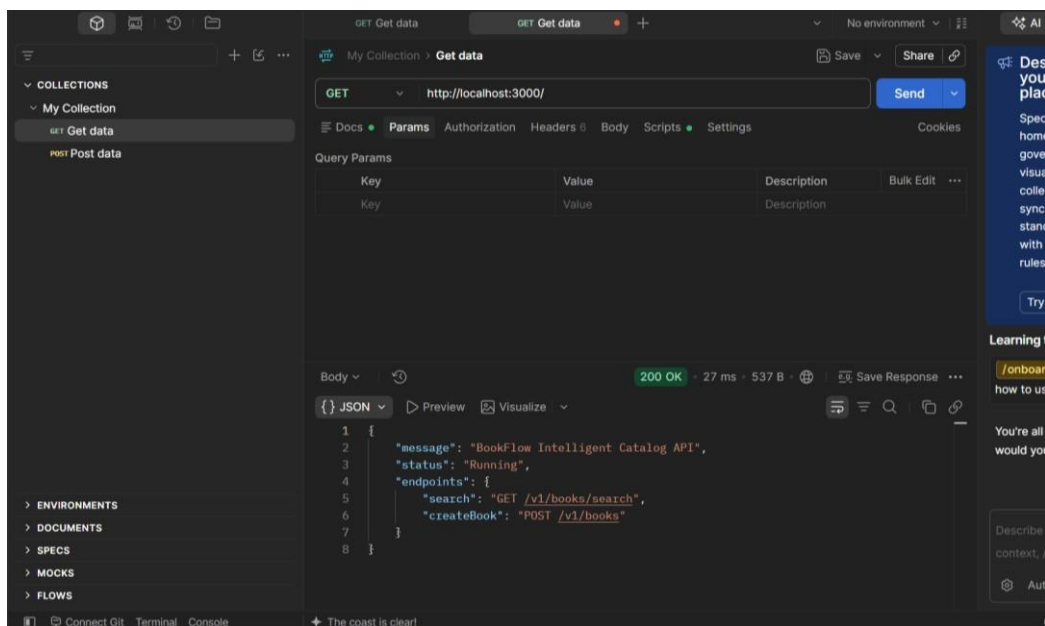
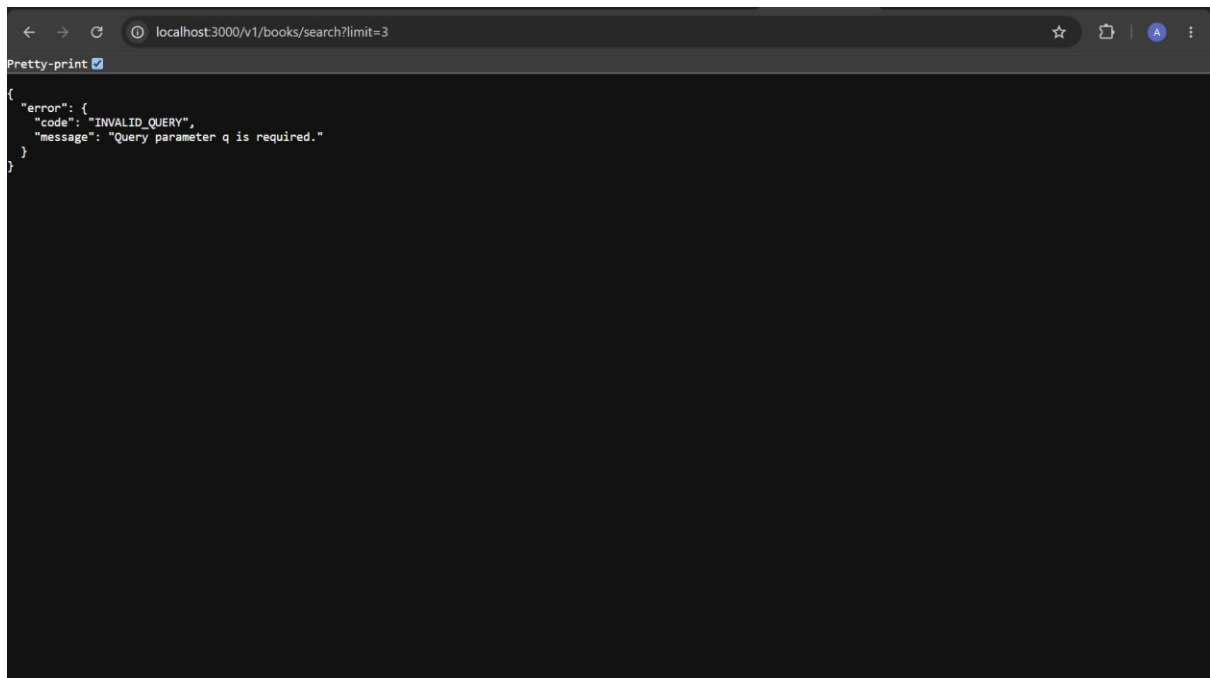
Action Output

#	Time	Action	Message	Duration / Fetch
23	14:56:26	SELECT book_id, title, author, genre, R...	3 row(s) returned	0.016 sec / 0.000 sec
24	15:03:59	SELECT book_id, title, author, genre, R...	3 row(s) returned	0.000 sec / 0.000 sec
25	21:30:47	SELECT book_id, title, author, genre, publi...	1 row(s) returned	0.110 sec / 0.000 sec

Object Info Session

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.



GET Get data

My Collection > Get data

GET http://localhost:3000/v1/books/search?q=Stories%20about%20space&limit=3 Send

Params Authorization Headers Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
q	Stories%20about%20space		
limit	3		

Body

200 OK · 78 ms · 1.01 KB Save Response

```
{
  "query": "Stories about space",
  "count": 3,
  "results": [
    {
      "book_id": 1,
      "title": "The Martian",
      "author": "Andy Weir",
      "genre": "Science fiction",
      "description": "An astronaut struggles to survive alone on Mars.",
      "published_year": 2011,
      "similarity": 1
    },
    {
      "book_id": 2
    }
  ]
}
```

The coast is clear!

GET Get data

My Collection > Get data

GET http://localhost:3000/v1/books/search?limit=3 Send

Params Authorization Headers Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
limit	3		

Body

400 Bad Request · 7 ms · 481 B Save Response

```
{
  "error": {
    "code": "INVALID_QUERY",
    "message": "Query parameter q is required."
  }
}
```

My Collection > Get data

GET http://localhost:3000/v1/books/search?q=space&limit=100

Send

Docs Params Authorization Headers 6 Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
✓ q	space		
✓ limit	100		

Body

400 Bad Request · 11 ms · 482 B

Save Response

JSON Preview Debug with AI

```
1 {
2   "error": {
3     "code": "INVALID_LIMIT",
4     "message": "limit must be between 1 and 50."
5   }
6 }
```

ENVIRONMENTS DOCUMENTS SPECS MOCKS FLOWS

Connect Git Terminal Console

Add response time and JSON validation tests

My Collection > Post data

POST http://localhost:3000/v1/books

Send

Docs Params Authorization Headers 8 Body Scripts Settings Cookies

raw JSON Schema Beautify

```
1 {
2   "title": "Project Hail Mary",
3   "author": "Andy Weir",
4   "genre": "Science Fiction"
5 }
```

Body

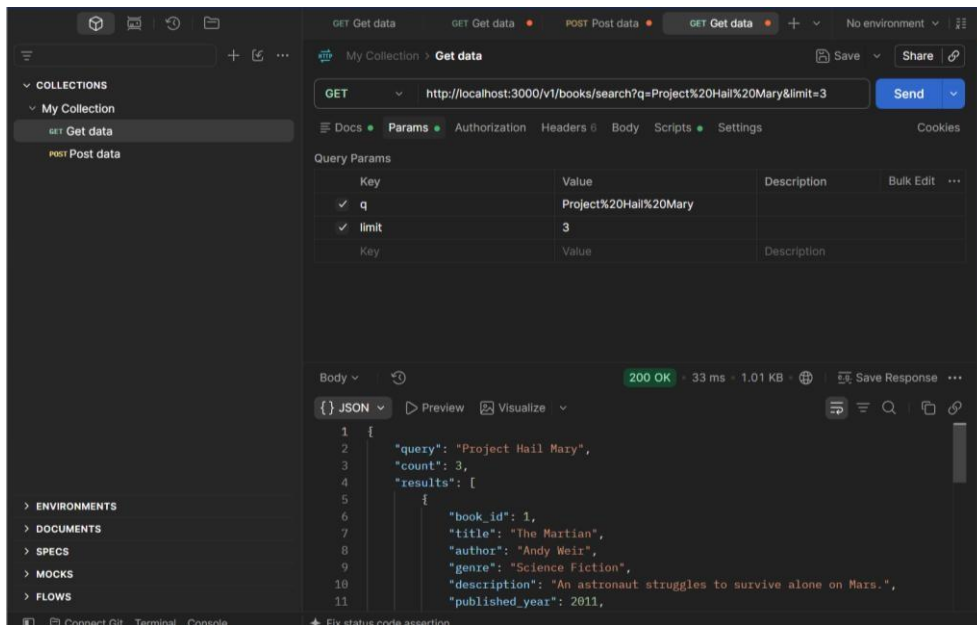
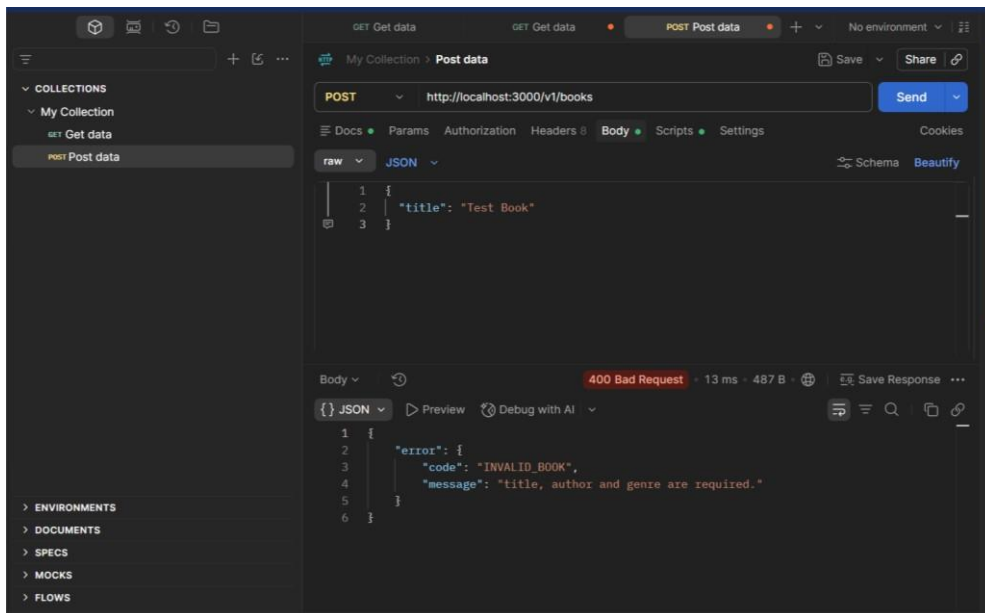
201 Created · 146 ms · 595 B

Save Response

JSON Preview Visualize

```
2 "message": "Book created successfully",
3 "book": {
4   "book_id": 12,
5   "title": "Project Hail Mary",
6   "author": "Andy Weir",
7   "genre": "Science Fiction",
8   "description": null,
9   "published_year": null,
10  "embedding": null
11 }
12 }
```

ENVIRONMENTS DOCUMENTS SPECS MOCKS FLOWS



MySQL Workbench

MySQL Model x Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator Week_1_practical SQL File 3* SQL File 4* x SQLAdditions

MANAGEMENT

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INSTANCE

- Startup / Shutdown
- Server Logs

Administration

Information

No object selected

Limit to 1000 rows

```
320 book_id,  
321 title,  
322 author,  
323 genre,  
324 published_year,  
325 embedding  
326 FROM Books  
327 ORDER BY book_id;  
328
```

Result Grid

book_id	title	author	genre	published_year
7	The Hobbit	J.R.R. Tolkien	Fantasy	1937
8	Enders Game	Orson Scott Card	Science Fiction	1977
9	Harry Potter and the Sorcerers Stone	J.K. Rowling	Fantasy	1997
10	Twenty Thousand Leagues Under the Seas	Jules Verne	Adventure	1870
11	Project Hail Mary	Andy Weir	Science Fiction	2020
12	Project Hail Mary	Andy Weir	Science Fiction	2020
* NULL	NULL	NULL	NULL	NULL

Books 9 x Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 24	15:03:59	SELECT book_id, title, author, genre, R...	3 row(s) returned	0.000 sec / 0.000 sec
✓ 25	21:30:47	SELECT book_id, title, author, genre, publi...	1 row(s) returned	0.110 sec / 0.000 sec
✓ 26	22:53:36	SELECT book_id, title, author, genre, publi...	12 row(s) returned	0.015 sec / 0.000 sec

Object Info Session

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.